

Multiple Choice (10 marks)

1. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 42.58 MHz
 - (c) 74.34 kHz
 - (d) 13.93 MHz
2. Calculate the magnetic field responsible for the polarization of 2.5×10^{-6} for ^{13}C at 298 K.
 3. 0.5 T
 4. 1.2 T
 5. 2.9 T
 6. 100 T

When the following equation is balanced, which of the following is true? $\text{MnO}_4^- + \text{I}^- + \text{H}^+ \rightleftharpoons \text{Mn}_2\text{O}_3 + \text{IO}_3^- + \text{H}_2\text{O}$

- (a) The $\text{I}^- : \text{IO}_3^-$ ratio is 3:1.
- (b) The $\text{MnO}_4^- : \text{I}^-$ ratio is 6:5.
- (c) The $\text{MnO}_4^- : \text{Mn}_2\text{O}_3$ ratio is 3:1.
- (d) The $\text{H}^+ : \text{I}^-$ ratio is 2:1.

The ^{13}C spectrum of which isomer of C_6H_{14} has lines with three distinct chemical shifts?

- (a) hexane
- (b) 2-methylpentane
- (c) 3-methylpentane
- (d) 2,3-dimethylbutane

Of the following ionic substances, which has the greatest lattice enthalpy?

- (a) MgO
- (b) MgS
- (c) NaF
- (d) NaCl

True / False (10 marks)

Answer True or False

6. True or False: 0.100 M NaCl has a higher ionic strength than 0.050 M AlCl₃ because it has more total solute concentration.
7. True or False: Halogenated hydrocarbons are safe and non-toxic pollutants in aquatic environments.
8. True or False: The polarization of a proton is 4.126×10^{-3} at 0.335 T and 2.142×10^{-6} at 10.5 T.
8. True or False: The ⁵⁵Mn nuclide has three allowed energy levels for its nuclear states.
9. True or False: The first ionization energy of argon is lower than that of neon due to its larger atomic size and electron shielding effects.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the procedures that can minimize the influence of random errors in measured results in chemistry, focusing on signal averaging and averaging results from multiple samples.
12. Discuss the factors influencing the EPR spectrum of a rigid nitronyl nitroxide diradical, specifically analyzing why nine lines would appear when $J \ll a$.

End of Paper